

Xiaozhen Ma

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EDUCATION

Southwest Jiaotong University and University of Leeds
2027

Sep 2023 - Jul

Bachelor of Engineering in **Electronic and Electrical Engineering** (Joint Program, dual degree from both universities)

ADDITIONAL EDUCATION AND ACADEMIC PROGRAM

University of California, Irvine
2026

Jun 2026 - Aug

UCInspire Summer Research Programme

St Catharine's College, Cambridge
2025

Feb

Interdisciplinary Winter School (Computer Science, Data Science, and Artificial Intelligence Track)

RESEARCH INTERESTS

Computer Vision, 3D Reconstruction, Robotic Perception, Autonomous Driving

RESEARCH EXPERIENCES

PIKAN-Drive: Progressive Physics-Informed KANs from PilotNet Steering Regression to Fast-BEV 3D Detection

Project Lead, Supervised by Prof. Fadi Kurdahi, UC Irvine

Jun 2026 - Present

- Led a progressive study of MLP, KAN, and PI-KAN architectures from PilotNet steering regression to Fast-BEV multi-camera 3D detection, establishing unified training, evaluation, and ablation pipelines.
- Built a PilotNet benchmark on 45,406 sequential Udacity samples; the best PI-KAN achieved MAE 3.209 / RMSE 6.163, reducing error by 11.3% / 9.7% relative to the MLP baseline, while the Fast-BEV KAN baseline improved mAP from 0.2559 to 0.2720 and NDS from 0.3950 to 0.4046.
- Designed nine grouped physics-informed auxiliary losses and conducted targeted ablations for Fast-BEV; showed that additional constraints did not consistently improve detection, motivating task-specific, metric-aligned, and architecture-aware supervision.

Intelligent CAD Generation and Point-Cloud-to-Editable-CAD Reconstruction

Research Assistant, Supervised by Prof. Dong-Ming Yan and Prof. Mingyang Zhao, National Laboratory of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences

Feb 2026 - Present

- Deployed and evaluated CAD-Assistant for tool-augmented CAD generation through the FreeCAD Python API, analysing task planning, code generation, geometric modelling, and execution feedback; optimised prompts with dimensional and geometric constraints to improve code executability, solid validity, and parametric modelling quality.
- Reproduced Point2CAD and ParSeNet, implemented data-interface compatibility between the two pipelines, and investigated multi-view and three-axis projection boundary constraints, showing that additional boundary supervision did not reliably improve the official segmentation baseline through comparative experiments.
- Developed a CAD-Recode-based point-cloud-to-parametric-CAD pipeline, generating executable CadQuery programs from point-cloud inputs, executing and validating them through CadQuery and OpenCascade/OCF, and producing editable CAD programs together with B-Rep / STEP / Solid outputs.
- Designed and evaluated CAD-knowledge supervision, geometric verification, parameter refinement, structural diagnosis, and multi-candidate selection strategies; the independently confirmed S2@100 inference pipeline reduced median Chamfer Distance by 13.4% relative to S0@10 while also improving mean, P90, and mean IoU without reducing solid validity, establishing geometric candidate reranking as the strongest confirmed approach.

Sparse Point Cloud 3D Railway Scene Reconstruction Technology Driven by Monocular Depth Estimation

Project Lead, Supervised by Prof. Zhao Tang, State Key Laboratory of Rail Transit Vehicle System

Apr 2025 - May

2026

- Developed the FrustumDense-SAM-XC3D railway-scene 3D reconstruction system by integrating multi-frame LiDAR point clouds, RGB images, and monocular depth information into a pipeline covering point-cloud densification, depth fusion, object segmentation, occlusion handling, and scene visualization.
- Proposed a multi-frame frustum back-projection densification method that projected LiDAR points from the current, historical, and future frames into the camera frustum for spatial aggregation, filling sparse regions around buildings, tracks, and trackside facilities to improve point-cloud coverage and structural completeness in large-scale scenes.
- Integrated DepthPro-based monocular metric depth estimation with LiDAR pose constraints and applied patch-wise fitting, boundary resampling, and moving least-squares prediction to correct scale errors in distant regions of the monocular depth maps and generate pixel-level dense 3D point clouds.
- Applied the pretrained SAM3 model to multi-frame image segmentation and object-boundary detection, used a tree-based data structure to manage nested objects and occlusion relationships, and implemented cross-frame co-visibility consistency checks for point-cloud matching, clustering, and unique object identification.
- Designed class-aware radius filtering and patch-based rendering algorithms to suppress accumulated noise from multi-

frame point-cloud fusion, extract tracks, buildings, and transportation facilities by category, and visualize large-scale railway scenes for single-frame object extraction and railway asset presentation.

Machine Learning-Driven 3D Reconstruction of High-Speed Train Head Shapes

Project Lead, Supervised by Prof. Zhao Tang, State Key Laboratory of Rail Transit Vehicle System
2025

Apr 2024 - Mar

- Constructed a portable point cloud acquisition device incorporating multi-sensor fusion of LiDAR, cameras, and IMUs. Developed a radar-vision tightly coupled SLAM framework based on the open-source FAST-LIVO2 algorithm, with extended Kalman filtering employed to optimise camera pose calculations.
- Employed an enhanced ICP algorithm to refine registration accuracy, with a dynamic threshold denoising technique. Generated uniform, high-precision point clouds through normal vector estimation and point cloud resampling.
- Introduced an incremental k-d tree (IK-D-Tree) to enhance map storage and retrieval efficiency, and a dynamic object exclusion algorithm (based on motion consistency detection) to improve localisation robustness.
- Proposed a Tri-axial Projection Segmentation Interpolation and Reverse Hole Boundary Identification Method to achieve point cloud densification and uniform distribution in sparse point cloud scenarios caused by regional hole boundaries, elevating model integrity from 75% to 98%.
- Explored adaptive mechanisms for multiple algorithms in complex surface modelling, including Greedy triangulation, Poisson reconstruction, B-spline interpolation, Moving cuboid algorithm, and Radial basis function (RBF) interpolation.
- Independently authored the research report titled “Three-Dimensional Surface Reconstruction of High-Speed Train Noses from Sparse Point Clouds” based on the project, systematically summarizing the technical approach and comparing experimental results.

PROJECT EXPERIENCES

FPGA-Based Digital System Design

May 2026 - Jun

2026

- Developed functional modules for a hierarchical digital system using Quartus and a hardware description language. Connected modules through predefined ports, performed logic simulation, waveform analysis, and FPGA hardware testing.

STM32-Based Embedded Game Development

Apr 2026 - Jun

2026

- Collaborated in a three-person team to develop an embedded game using STM32 hardware and peripherals, contributing core game mechanics and an independent feature or level. Managed code with GitHub and documented development logs, system tests and demonstrations.

Audio-Visualisation PCB Design and Fabrication

Mar 2026 - May

2026

- Created the schematic and PCB layout for an audio-visualisation circuit in KiCad, using LEDs to indicate low-, mid- and high-frequency signals. Soldered electronic components, and used a multimeter for measurement and troubleshooting.

Emerging Optoelectronic System Design

Oct 2025 - Mar

2026

- Worked in an international team with University of Leeds students to research and design an optoelectronic system addressing a space-related challenge. Analysed its technical principles, commercial applications, environmental impact and engineering ethics, and produced an English-language design presentation video.

MCU Embedded Systems: Line-Following Model Car & Washing Machine Controller

Apr 2025 - May

2025

- Performed hardware programming of the line-following model car using Keil5, implementing functions such as LED display based on the STM32 microcontroller. Constructed a control circuit using MOSFETs as switches to control the model car's movement and turning. Designed and handcrafted the model car's body.
- Programmed the washing machine controller using Mbed to implement functions such as mode selection, child lock, time selection, pressure response, and light sensitivity based on the STM32 microcontroller.

HONORS AND AWARDS

- Second Prize (AI Innovation Track), 27th China Robot & Artificial Intelligence Competition - National Finals, 2025. (An Online Sports Course Evaluation System Based on Multi-Expert Collaborative Systems and Sentiment Analysis)
- Outstanding Student Leader, SWJTU-Leeds Joint School, Southwest Jiaotong University, 2025.
- Mingcheng Award, SWJTU-Leeds Joint School, Southwest Jiaotong University, 2024.

EXTRACURRICULAR ACTIVITIES AND VOLUNTEER

Academic Innovation Base of SWJTU-Leeds Joint School, Southwest Jiaotong University

Sep 2023 - Present

Roles: Member (2023-2024); Head of Activities (2024-2025); President (2025-Present)

- As president, coordinated over 30 members and 4 departments, liaised with more than 80 guests, planned event schedules, venues, and themes, and spearheaded the development of the “Leeds Tech Month” flagship competition, attracting over 500 participating teams, with events reaching more than 3,000 faculty and students.

TECHNICAL SKILLS AND ADDITIONAL INFORMATION

Technical Skills: C/C++/Python/MATLAB(VS2017, MbedStudio, OnlineGDB, Keil5); ROS2; Linux; SLAM (FAST-LIO/FAST-LIVO2/R3LIVE); point-cloud/3D reconstruction; VTK/PCL/OpenCV; embedded integration (STM32).

Languages: Mandarin(Native), English(Fluent). **Interests:** Eight Brocades, Simplified 24-Form Tai Chi Chuan